(iii) (a) $P(X \geq 4)$

$$= P(X = 4) + P(X = 5) + P(X = 6)$$

$$= 9a + 11a + 13a$$

$$= 33a$$

$$= 33 \times \frac{1}{49} = \frac{33}{49}$$

(b) $P(X<5)$ $= 1 - P(X \geq 5) = 1 - [P(X = 5) + P(X = 6)]$

$$= 1 - [11a + 13a]$$

$$= 1 - 24a$$

$$= 1 - \frac{24}{29}$$

$$= \frac{25}{29}$$

(c) $P(3 \leq x \leq 6)$ $= P(X = 3) + P(X = 4) + P(X = 5) + P(X = 6)$

$$= 7a + 9a + 11a + 13a$$

$$= 40$$

$$a = \frac{40}{49}$$

(iv) $P(X = 5)$ $= F(5) - F(5 - 1)$

$$= \frac{36}{49} - \frac{25}{49}$$

$$= \frac{11}{49}.$$

Example 9.6

Let X be a random variable with p.d.f

$$f(x) = \begin{cases} \frac{x}{2}; & 0 < x < 2 \\ 0 \ ; & otherwise \end{cases}$$

(i) Find the c.d.f of X , (ii) Compute $P\left(\frac{1}{2} < X \leq 1\right)$, (iii) Compute $P(X=1.5)$

Solution:

(i)The c.d.f of X : $F(x) = P(X \leq x)$

$$= \int_{-\infty}^{x} f(x)dx$$